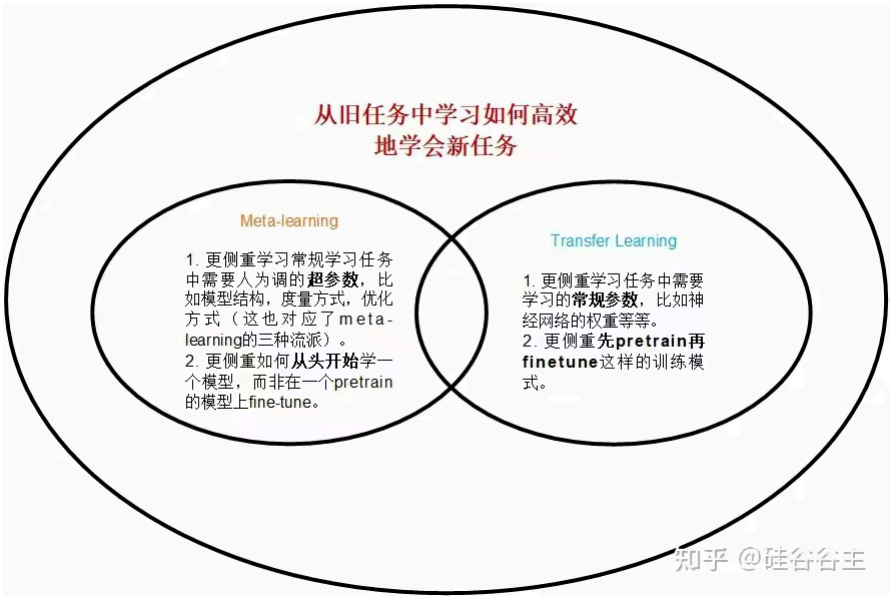
11.25 Transfer Learning方式

- 两种思路：Regualarization+DRO, 一定可行；Mixture Distribution

Multi-Retailer: 牵涉到Meta-Learning, Personalized Federated Learning 联合学习，相当于transfer learning的泛化版本。

区别在于

1. Transfer learning 只调优一个任务的weight，注重常规参数;
2. **meta-learning adapts to many new tasks efficiently** 快速迁移，而且更注重宏观模型meta model;
3. PFL额外关注**privacy**，每个user只接收其他模型的weight，而非整体data



- **Transfer Learning**: 以Proxy为样本，对Gold Data调优，**适用范围：只考虑整体对局部的要求，而局部不影响整体。**

  同时对多个任务进行Regularized local loss，则为meta learning.

- STEP 1: OLS estimator of Proxy Data

$$\hat{\beta}_{proxy} = \arg\min_{\beta} \left\{ \frac{1}{n_{proxy}} \|\mathbf{Y}_{proxy} - \mathbf{X}_{proxy}\beta\|_2^2 \right\}$$

- STEP 2: LASSO estimator of Gold Data debiased by Proxy Estimator

$$\hat{\beta}_{joint}(\lambda) = \arg\min_{\beta} \left\{ \frac{1}{n_{gold}} \|\mathbf{Y}_{gold} - \mathbf{X}_{gold}\beta\|_2^2 + \boxed{\lambda \|\beta - \hat{\beta}_{proxy}\|_1} \right\} \cdot \textbf{Regularization}$$

When $\lambda = 0$, it is equivalent to OLS estimator of Gold Data $\quad \boldsymbol{\beta}^* = \boldsymbol{\beta}_j^* + \delta_j^*, \forall j \in [J]$

- **Personalized Federate Learning （PFL)/Meta Learning**: 在上述基础上，第一步的proxy优化是知道第二步的优化参数y_i，最优化整体的loss。**优点：整体影响局部，而局部又影响整体**
  - Personalized Federated Learning with Moreau Envelopes

Meta-Learning and Personalized Federated Learning. Both applications can be viewed asspecial cases of CSBO. For meta-learning with M tasks or personalized federated learning with M users, the goal is to find a common regularization center θ shared by all tasks or all users.

$$\min_{x} \ \mathbb{E}_{i\sim\mu}\mathbb{E}_{D_i^{test}\sim\rho_i}\left[l_i(y_i^*(x), D_i^{test})\right] \qquad \text{(upper level)}$$

$$\text{where } y_i^*(x) = \text{argmin}_{y_i} \ \mathbb{E}_{D_i^{train}\sim\rho_i}\left[l_i(y_i, D_i^{train}) + \frac{\lambda}{2}\|y_i - x\|^2\right], \forall i \in [M], x. \qquad \text{(lower level)} \tag{4}$$

Here, μ is the empirical uniform distribution on $[M]$. The upper-level problem minimizes the generalization loss for all tasks/all users by tuning the joint regularization center x, and the lower-level problem finds an optimal regularization parameter x_i close to x for each individual task or user. Note that M may be as large as $\mathcal{O}(10^3)$ in meta-learning and as large as $\mathcal{O}(10^9)$ in personalized federated learning. Thus, it is crucial to design methods with complexity bounds independent of M.

  - **Bi-level Optimization**: The upper level problem minimizes **the generalization loss for all tasks/all users** by **tuning the joint regularization center x**；

    and the lower level problem finds an optimal regularization parameter x_i close to x for each individual task or user.

  - PFL对n个客户协同训练不同的模型，**每个retailer只保留自己的数据**，对应参数为θ_i，总的目标是最小化总loss function:

$$\Theta^* = \arg\min_{\Theta} \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}_{\boldsymbol{x},y\sim\mathcal{P}_i}\left[\ell_i(\boldsymbol{x}_j, y_j; \theta_i)\right] \tag{1}$$

    每个loss function定义为：

$$\arg\min_{\Theta} \frac{1}{n} \sum_{i=1}^{n} \mathcal{L}_i(\theta_i) = \arg\min_{\Theta} \frac{1}{n} \sum_{i=1}^{n} \frac{1}{m_i} \sum_{j=1}^{m_i} \ell_i(\boldsymbol{x}_j, y_j; \theta_i)$$

- 处理PFL的几种方式：重点参考mixing model, regularization和local fine tuning

  1. **Mixing models /Model interpolation** 将local model和global model混合：local data数量少，希望利用global model的泛化性，同时降低heterogeneity造成性能下降；

     Adaptive Personalized Federated Learning

     - Global model: $\bar{h}^* = \arg\min_{h\in\mathcal{H}} \hat{\mathcal{L}}_{\bar{\mathcal{D}}}(h)$，其中$\bar{\mathcal{D}} = (1/n)\sum_{i=1}^{n}\mathcal{D}_i$，平均成本最低；
     - Local model: 考虑部分global model； 这里结合使用了Weighted Loss Estimator

$$\hat{h}^*_{loc,i} = \arg\min_{h\in\mathcal{H}} \hat{\mathcal{L}}_{\mathcal{D}_i}(\alpha_i h + (1-\alpha_i)\bar{h}^*),$$

     - 最终estimator为global和local的**凸组合**，Model Averaging Estimator已被证明效果不好

$$h_{\alpha_i} = \alpha_i \hat{h}^*_{loc,i} + (1-\alpha_i)\bar{h}^*,$$

     Federated Learning of a Mixture of Global and Local Models
     另一种方法：类似于Regularization，只不过一次优化，只考虑训练一个FL模型，联合优化所有local参数。Bastani考虑的也是一个mixed model，只不过分成两部。

We now introduce our new formulation for training supervised FL models:

$$\min_{x_1,\ldots,x_n\in\mathbb{R}^d} \{F(x) := f(x) + \lambda\psi(x)\}$$
$$f(x) := \frac{1}{n}\sum_{i=1}^{n} f_i(x_i), \quad \psi(x) := \frac{1}{2n}\sum_{i=1}^{n}\|x_i - \bar{x}\|^2, \tag{2}$$

where $\lambda \geq 0$ is a penalty parameter, $x_1,\ldots,x_n \in \mathbb{R}^d$ are local models, $x := (x_1, x_2, \ldots, x_n) \in \mathbb{R}^{nd}$ and $\bar{x} := \frac{1}{n}\sum_{i=1}^{n} x_i$ is the average of the local models.

     启发：假设global model的residual distribution为$\mathbb{P}$, local model的是$\mathbb{P}_i$，mixture distribution对应的集合为分布凸组合，在此基础上增加robust；mixture的程度影响personalization效果

  2. **Meta-learning / Local fine-tuning:**通过Model-Agnostic Meta-Learning (MAML)， 先学一个global model再fine tune，目标函数是最小化fast adaption，有梯度；

Personalized Federated Learning: A Meta-Learning Approach

Improving federated learning personalization via model agnostic meta learning

$$\min_{\mathbf{w}\in\mathbb{R}^d} F(\mathbf{w}) = \frac{1}{n}\sum_{i=1}^{n}\mathcal{L}_i(\mathbf{w}-\alpha\nabla\mathcal{L}_i(\mathbf{w})),$$

Bilevel Optimization: 其中$\boldsymbol{w}$是meta-model，而gradient代表从meta-model进行adaptation，更改梯度

启发：类似于Transfer learning，局部调优，建立集合；这种方法缺点是overfit，应该考虑泛化性能

3. **Model Regularization**: 对local model采取regularized loss，对global和local之间的距离采用regularization

   Personalized Federated Learning with Moreau Envelopes

   $$f_i(\theta_i) + \frac{\lambda}{2}\|\theta_i - w\|^2,$$

   与Bastani不同的是，总目标采用Bilevel Problem，上层问题考虑下层问题影响：

   $$\texttt{pFedMe}: \min_{w\in\mathbb{R}^d}\Big\{F(w) := \frac{1}{N}\sum\nolimits_{i=1}^{N} F_i(w)\Big\}, \text{ where } F_i(w) = \min_{\theta_i\in\mathbb{R}^d}\Big\{f_i(\theta_i) + \frac{\lambda}{2}\|\theta_i - w\|^2\Big\}.$$

   启发：用global和local的远近，对模型进行regularize；是否加入bilevel optimization

4. **Clustering** 先将相似user聚类，再对每个group训练一个model，只有相似的subset有共同模型.

   Personalized federated learning with first order model optimization.

   Three approaches for personalization with applications to federated learning.

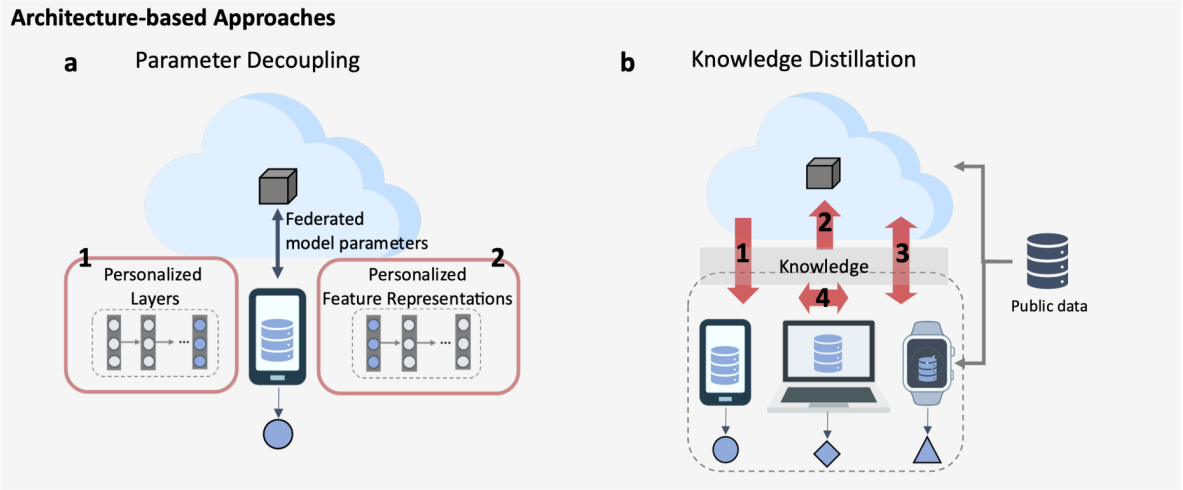
5. **Multi-task Relationship learning**: 和Meta learning类似，训练一个模型可以在多个任务表现优良，然后根据不同task的pair-wise Relationship区别（例如Covariance Matrix)，进一步对问题调优。

- **STEP 1**： global model是**linear combination of local model**，和single global model不同

$$u_c = \xi_{c,1}\theta_1 + \ldots + \xi_{c,m}\theta_r$$

- **STEP 2**: Personalized model进行调优

$$\theta_c^* = \arg\min_{\theta \in \mathbb{R}^d} f_c(\theta) + \frac{\mu}{2\alpha}\|\theta - u_c\|^2$$

6. **Separable Model / Parameter Decoupling**：将模型的parameter做切分，global model只关注自己部分的训练，local model训练personalized。类似于[需求定价可分模型]；Split learning, 其中feature可以切分，每种feature选择不同模型，相当于训练不同部分，再组合起来。

7. **Knowledge Distillation**：知识蒸馏，将现存模型的知识进行浓缩，然后传递；根据方向不同，可分为4类